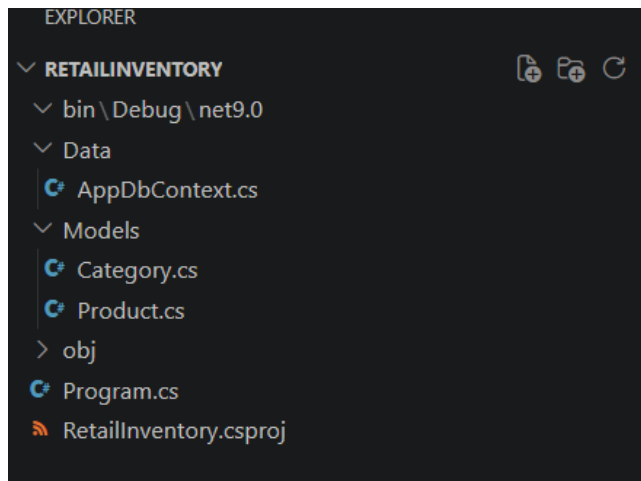


Lab 2: Setting Up the Database Context for a Retail Store

Aim:

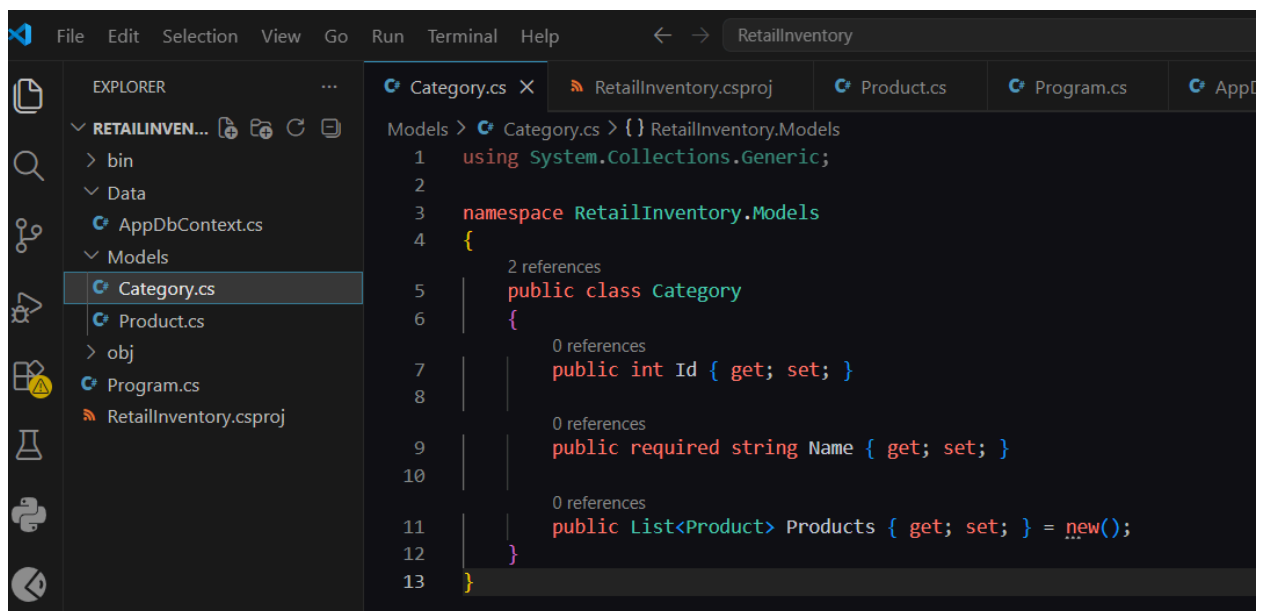
To configure **Entity Framework Core DbContext**, create entity models, and establish a connection between a .NET Console Application and SQL Server.

Project Structure:

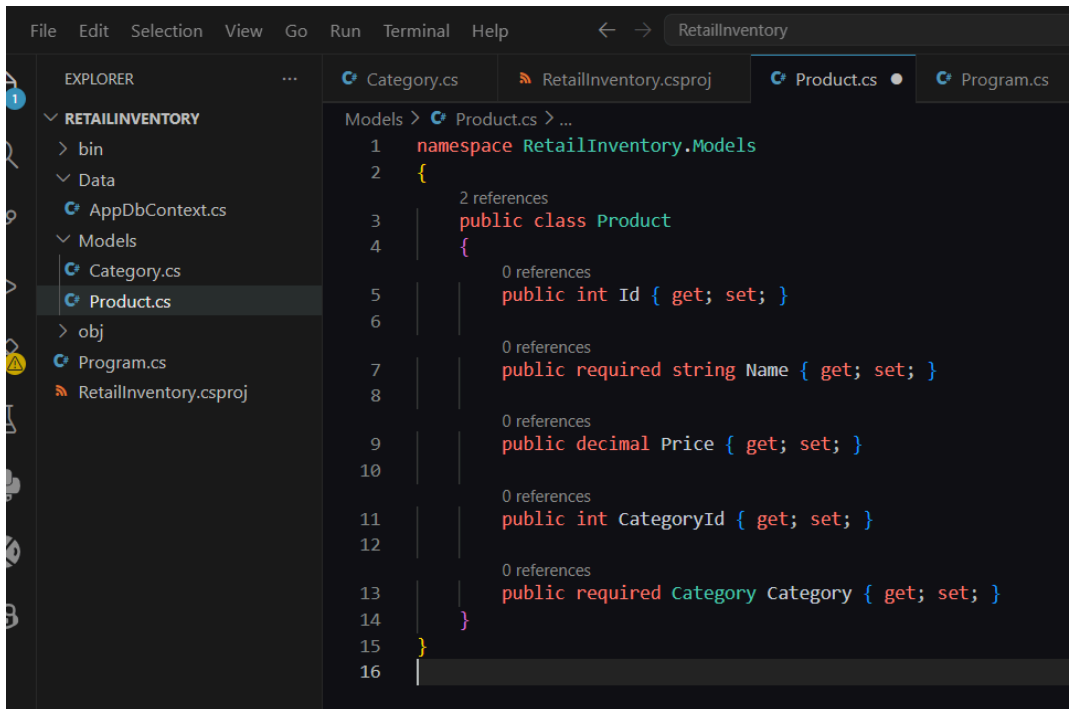


1.Models

[Category.cs](#)



[Product.cs](#)

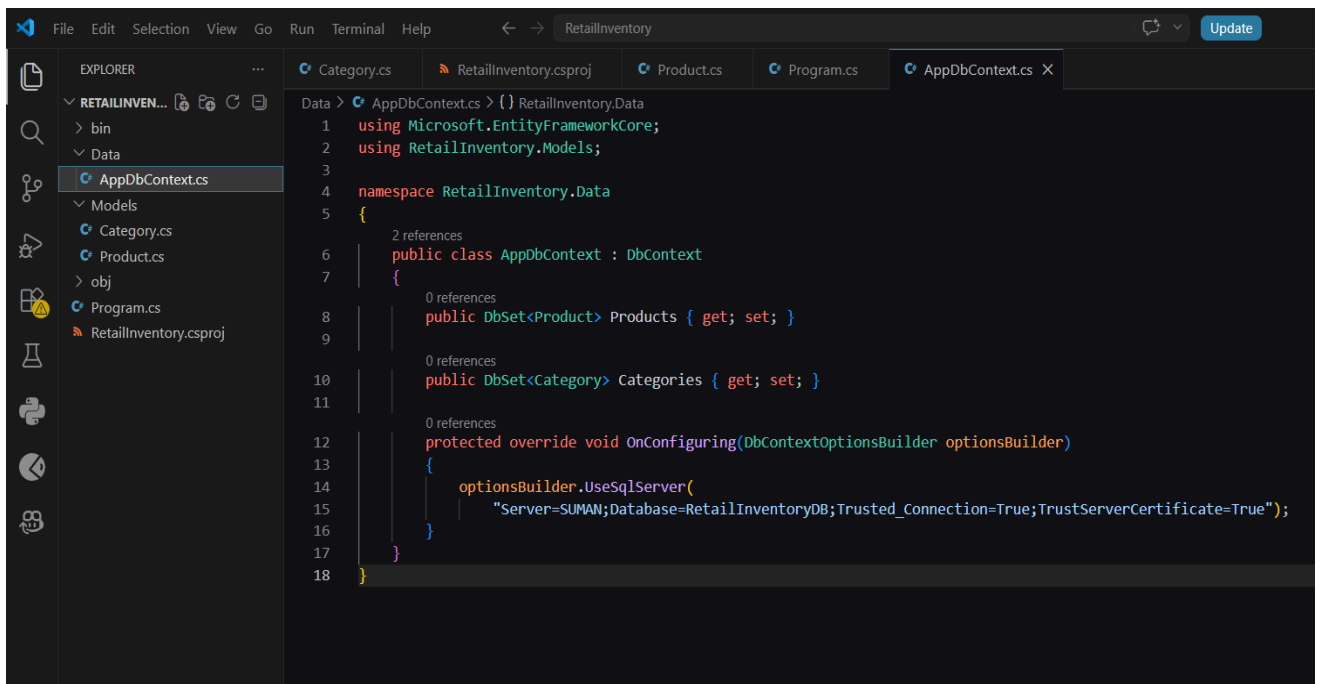


The screenshot shows the Visual Studio IDE with the 'RetailInventory' project open. The 'EXPLORER' pane on the left shows the project structure: 'bin', 'Data', 'AppDbContext.cs', 'Models' (containing 'Category.cs' and 'Product.cs'), 'obj', 'Program.cs', and 'RetailInventory.csproj'. The 'Product.cs' file is selected in the 'Models' folder. The main editor displays the code for 'Product.cs' with line numbers 1 through 16. The code defines a namespace 'RetailInventory.Models' and a class 'Product' with properties 'Id', 'Name', 'Price', 'CategoryId', and 'Category'.

```
1 namespace RetailInventory.Models
2 {
3     2 references
4     public class Product
5     {
6         0 references
7         public int Id { get; set; }
8
9         0 references
10        public required string Name { get; set; }
11
12        0 references
13        public decimal Price { get; set; }
14
15        0 references
16        public int CategoryId { get; set; }
17
18        0 references
19        public required Category Category { get; set; }
20    }
21 }
```

2. Create AppDbContext

AppDbContext.cs



The screenshot shows the Visual Studio IDE with the 'RetailInventory' project open. The 'EXPLORER' pane on the left shows the project structure: 'bin', 'Data' (containing 'AppDbContext.cs'), 'Models' (containing 'Category.cs' and 'Product.cs'), 'obj', 'Program.cs', and 'RetailInventory.csproj'. The 'AppDbContext.cs' file is selected in the 'Data' folder. The main editor displays the code for 'AppDbContext.cs' with line numbers 1 through 18. The code defines a namespace 'RetailInventory.Data' and a class 'AppDbContext' that inherits from 'DbContext'. It includes properties for 'Products' and 'Categories', and an override for 'OnConfiguring' that sets up the database connection.

```
1 using Microsoft.EntityFrameworkCore;
2 using RetailInventory.Models;
3
4 namespace RetailInventory.Data
5 {
6     2 references
7     public class AppDbContext : DbContext
8     {
9         0 references
10        public DbSet<Product> Products { get; set; }
11
12        0 references
13        public DbSet<Category> Categories { get; set; }
14
15        0 references
16        protected override void OnConfiguring(DbContextOptionsBuilder optionsBuilder)
17        {
18            optionsBuilder.UseSqlServer(
19                "Server=SUMAN;Database=RetailInventoryDB;Trusted_Connection=True;TrustServerCertificate=True");
20        }
21    }
22 }
```